

FIT9132 Applied Week 4

A5-1 The Normalisation Process

A5-1.1 Data Anomalies

dentist_no	dentist_name	patient_no	patient_name	appointment		surgeryroom_no
				date	time	
D1011	Tony Smith	P100	Gillian White	12-Aug-2022	10:00	S15
D1011	Tony Smith	P105	Jill Bell	12-Aug-2022	12:00	S15
D1024	Helen Pearson	P108	Ian MacKay	12-Aug-2022	10:00	S10
D1024	Helen Pearson	P108	Ian MacKay	14-Aug-2022	14:00	S10
D1032	Robin Plevin	P105	Jill Bell	14-Aug-2022	16:30	S15
D1032	Robin Plevin	P110	John Walker	15-Aug-2022	18:00	S13

Insert Anomalies: If I need to insert a new dentist, I need to insert the dentist detail in DENTIST Table first before making changes in this table.

Delete Anomalies: If I need to delete a row in this table, I need to first delete Dentist details in the DENTIST Table

Update Anomalies: Before updating this table, I need to update the DENTIST or PATIENTS table first.

A5-1.2 The Normalisation Process

APPOINTMENT (dentist_no, dentist_name, patient_no, patient_name, app_datetime, surgeryroom_no)

UNF

APPOINTMENT (dentist_no, dentist_name, patient_no, patient_name, app_datetime, surgeryroom_no)

1NF

APPOINTMENT (dentist_no, app_datetime, dentist_name, patient_no, patient_name, surgeryroom_no)

CKs:

dentist_no, app_datetime

patient_no, app_datetime

surgeryroom_no, app_datetime

Partial dependencies:

dentist_no $\rightarrow$ dentist_name

patient_no $\rightarrow$ patient_name

*note that we use general definition, partial dependency is based on all candidate keys (PK and alternate keys)

2NF

APPOINTMENT (dentist_no, app_datetime, patient_no, surgeryroom_no)

DENTIST (dentist_no, dentist_name)

PATIENT (patient_no, patient_name)

Transitive dependencies:

None

3NF

There is no transitive dependency, the 3NF is the same as the 2NF. Note that you are required to show all forms, even if they are the same as a previous form.

APPOINTMENT (dentist_no, app_datetime, patient_no, surgeryroom_no)

DENTIST (dentist_no, dentist_name)

PATIENT (patient_no, patient_name)

Full Dependencies:

dentist_no, app_datetime → patient_no, surgeryroom_no

dentist_no → dentist_name

patient_no → patient_name